

Research Notes on Universal approximation theorem

Universal approximation theorem

$$f(x) = \sum_{i=1}^N v_i \sigma(w_i^T x + b_i)$$

>> Section 1

Quantitative bounds The question of minimal possible width for universality was first studied in 2021, Park et al obtained the minimum width required for the universal approximation of L_p functions using feed-forward neural networks with ReLU as activation functions. Similar results that can be directly applied to residual neural networks were also obtained in the same year by Paulo Tabuada and Bahman Ghahesifard using control-theoretic arguments. In 2023, Cai obtained the optimal minimum width bound for the universal approximation. For the arbitrary depth case, Leonie Papon and Anastasis Kratsios derived explicit depth estimates depending on the regularity of the target function and of the activation function.

>> Section 2

Arbitrary depth The arbitrary depth case was also studied by a number of authors such as Gustaf Gripenberg in 2003, Dmitry Yarotsky, Zhou Lu et al in 2017, Boris Hanin and Mark Sellke in 2018 who focused on neural networks with ReLU activation function. In 2020, Patrick Kidger and Terry Lyons extended those results to neural networks with general activation functions such, e.g. tanh or GeLU. One special case of arbitrary depth is that each composition component comes from a finite set of mappings. In 2024, Cai constructed a finite set of mappings, named a vocabulary, such that any continuous function can be approximated by composing a sequence from the vocabulary. This is similar to the concept of compositionality in linguistics, which is the idea that a finite vocabulary of basic elements can be combined via grammar to express an infinite range of meanings.

>> Section 3

Bounded depth and bounded width The bounded depth and bounded width case was first studied by Maiorov and Pinkus in 1999. They showed that there exists an analytic sigmoidal activation function such that two hidden layer neural networks with bounded number of units in hidden layers are universal approximators. In 2018, Guliyev and Ismailov constructed a smooth sigmoidal activation function providing universal approximation property for two hidden layer feedforward neural networks with fewer units in hidden layers. In 2018, they also constructed single hidden layer networks with bounded width that are still universal approximators for univariate functions. However,

this does not apply for multivariable functions. In 2022, Shen et al. obtained precise quantitative information on the depth and width required to approximate a target function by deep and wide ReLU neural networks.

>> Section 4

In the field of machine learning, the universal approximation theorems (UATs) state that neural networks with a certain structure can, in principle, approximate any continuous function to any desired degree of accuracy. These theorems provide a mathematical justification for using neural networks, assuring researchers that a sufficiently large or deep network can model the complex, non-linear relationships often found in real-world data. The best-known version of the theorem applies to feedforward networks with a single hidden layer. It states that if the layer's activation function is non-polynomial (which is true for common choices like the sigmoid function or ReLU), then the network can act as a "universal approximator." Universality is achieved by increasing the number of neurons in the hidden layer, making the network "wider." Other versions of the theorem show that universality can also be achieved by keeping the network's width fixed but increasing its number of layers, making it "deeper." These are existence theorems. They guarantee that a network with the right structure exists, but they do not provide a method for finding the network's parameters (training it), nor do they specify exactly how large the network must be for a given function. Finding a suitable network remains a practical challenge that is typically addressed with optimization algorithms like backpropagation.

>> Section 5

Arbitrary-depth case The "dual" versions of the theorem consider networks of bounded width and arbitrary depth. A variant of the universal approximation theorem was proved for the arbitrary depth case by Zhou Lu et al. in 2017. They showed that networks of width $n + 4$ with ReLU activation functions can approximate any Lebesgue-integrable function on n -dimensional input space with respect to L^1 distance if network depth is allowed to grow. It was also shown that if the width was less than or equal to n , this general expressive power to approximate any Lebesgue integrable function was lost. In the same paper it was shown that ReLU networks with width $n + 1$ were sufficient to approximate any continuous function of n -dimensional input variables. The following refinement, specifies the optimal minimum width for which such an approximation is possible and is due to.

>> Section 6

Here " $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ " is " λ -strictly

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increasing on some set X " means that there exists a strictly increasing function $u : X \rightarrow \mathbb{R}$ such that $|\sigma(x) - u(x)| \leq \lambda$ for all $x \in X$. Clearly, a λ -increasing function behaves like a usual increasing function as λ gets small. In the "depth-width" terminology, the above theorem says that for certain activation functions depth-2 networks are universal approximators for univariate functions and depth-3 networks are universal approximators for d -variable functions ($d > 1$).

>> Section 7

Arbitrary-width case A universal approximation theorem formally states that a family of neural network functions is a dense set within a larger space of functions they are intended to approximate. In more direct terms, for any function f from a given function space, there exists a sequence of neural networks $\phi_1, \phi_2, \dots$ from the family, such that $\phi_n \rightarrow f$ according to some criterion.

>> Section 8

A spate of papers in the 1980s—1990s, from George Cybenko and Kurt Hornik etc, established several universal approximation theorems for arbitrary width and bounded depth. See for reviews. The following is the most often quoted: Also, certain non-continuous activation functions such as the step function can be approximated by continuous activation functions, which then allows the approximation result to apply to those functions. In particular, this shows that a perceptron network with a single infinitely wide hidden layer can approximate arbitrary functions. Such an f can also be approximated by a network of greater depth by using the same construction for the first layer and approximating the identity function with later layers.